

C-STGB: Conformal Spatio-Temporal GraphBoost End-to-End Architecture

TIER 1: STREAMING INGEST

DuckDB Invariants & Hawkes

- **DuckDB / Arrow Engine**
 $O(1)$ zero-copy micro-batch
- **Dynamic Top-K Capping**
 $K \leq 15$ neighbor ceiling
- **12-D Invariants**
 $\mathbf{z}_{\text{inv}} \in \mathbb{R}^{12}$ flow basis
- **Hawkes Intensity**
 $\lambda_u(t)$ burst acceleration

Latency: 0.45 ms | DuckDB

TIER 2: GNN BACKBONE

Tri-Band & Camouflage Filter

- $\mathbf{z}_{\text{inv}}, \mathcal{G}$ →
- **Tri-Band Harmonic Attn.**
 $\Phi_{\text{time}}(\Delta t)$ harmonic bank
 - **Edge-Trust Gating**
 $g_{ij} \geq 0.10$ (65.9% pruned)
 - **Typology GraphSMOTE**
 $\mathcal{C}_k$ latent interpolation
 - **Hard-Negative Mining**
Suppresses merchant hubs

Continuous-Time HGT

TIER 3: FUSION & RISK

Evidence-Adaptive Bayes

- $\mathbf{h}_u^{(L)}$ →
- **Evidence-Adaptive Gate**
 $\alpha_u \in [0, 1]$ cold-start blend
 - **Cold-Start Embed Fused**
 $\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1 - \alpha_u) \mathbf{z}_{\text{inv}}$
 - **Asym. Focal Tversky Loss**
 $\mathcal{L}_{\text{task}}$ cost-sensitive
 - **Optimal Bayes Policy**
 τ^* threshold tuning

Cold-Start + Bayes Risk

TIER 4: CONFORMAL SWARM

CRC Triage & FinCEN SAR

- $\hat{p}_u, \mathbf{h}_u^*$ →
- **Class-Conditional CRC**
 $1 - \alpha \geq 99.0\%$ coverage
 - **3-Tier Selective Triage**
 $\Gamma(X) \subseteq \{0, 1\}$
 - **Multi-Agent Swarm**
AST compiler invariant
 - **FinCEN Form 111 XML**
SHA-256 Merkle audit seal

Fed SR 11-7 Verifiable